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(1) Research Program/Project/Study Title: Nodulation, Growth, and Yield Responses of Leguminous Crops Applied with Plant Growth Regulators (PGRs) Under Coffee-Based Farming System	
Proponent Name:	Ramonita C. Verano
Academic Rank:	Associate Professor I
Co-Authors (with academic rank):	Freddie L. Baranda – Assistant Professor III Ryan T. Sarimong – Associate Professor V Monalyn L. Oloroso - Associate Professor I Eddie P. Gacasan – Instructor I
Campus/Satellite College/Address:	Burias Campus
Telephone/Cell Phone/Fax Number:	0966-758-4854
E-mail Address:	rcverano@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation <table border="1" style="width: 100%;"> <tr> <td>Municipality/City:</td> <td>Mambusao</td> </tr> <tr> <td>Province:</td> <td>Capiz</td> </tr> <tr> <td>Region:</td> <td>VI</td> </tr> </table>				Municipality/City:	Mambusao	Province:	Capiz	Region:	VI
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(5) Classification <table border="1" style="width: 100%;"> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Coffee, Leguminous Crops		(8) Discipline/Field of Specialization Agriculture							

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The Philippines is one of the few countries that produces the four species of coffee: Arabica, Liberica (Barako), Excelsa and Robusta. Climatic and soil conditions in the Philippines - from the lowland to mountain regions - make the country suitable for all four species (<https://www.commune.ph/100-philippine-coffee/>).

Small farmers are the country's main producers of coffee in Mindanao. Robusta accounts for 66% of coffee produced, 25% for Arabica, 8% for Excelsa, and 1% for Liberica. However, the country's coffee supply is not enough, with coffee production supply significantly at a down trend from 72,342 MT DC (36,171 MT GCB) in 2015 to 60,640.95 MT DC (30,320.47 MT GCB) in 2020. Its yield averaged of 0.64 MT/ha 2015 declined to 0.54 MT/ha in 2020 (Department of Agriculture - Bureau of Agricultural Research, 2022). Among the three Visayan regions, Western Visayas has the biggest coffee production with a potential crop volume of 400 MT, based DA records (Conserva, 2017).

Coffee plays a significant role in Philippine culture and society. It is a staple beverage consumed throughout the day, from breakfast to after-dinner conversations. Coffee shops have become a popular gathering place for friends and colleagues, with many local coffee shops offering unique blends and flavors.

Improved agricultural technologies were introduced in the whole parts of the country including coffee-based farming systems through agricultural research systems, universities, agriculture offices, NGOs and have been promoted to the farmers to address problems related to production and productivity in a sustainable manner (Girma et al., 2022). Technologies including conservation agriculture, organic fertilizers, improved crop varieties, improved livestock and improved feeding practices, and soil and water conservation practices have been promoted to farmers by stakeholders like the Ministry of agriculture, research and NGOs (CIAT, 2017).

Coffee-based farming systems (CBFS) support smallholder farmers through mainly coffee growing with integration of other food crops and livestock. Climate change is expected to ravage crop suitability in several agroecological zones, posing a threat to national earnings and livelihoods. However, previous studies have mainly considered crop-specific analyses rather than the major crops in a farming system.

Plant growth regulators when applied in very small quantity influence the plant growth. Several reports indicated that application of growth regulator improved the plant growth and yield of several plants (Hernandez, 1997). It plays an important role in plant growth and development, but little is known about the roles of plant growth regulators in the yield components and seed qualities of leguminous crops. It is therefore considered imperative to study the effect of plant growth regulators on the nodulation, growth and yield attributes of soybean, peanut and mung bean.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The project will focus on the concept on nodulation, growth, and yield Responses of Leguminous Crops Applied with different plant growth regulators (PGRs) Under Coffee-Based Farming System. Some efforts had been made to study adoption of agricultural technologies in coffee-based farming system though these studies were either location specific or focused on single technology adoption and did not consider the nature of simultaneity of adoption decision among agricultural technologies.

Nevertheless, cover crops are known to be a sustainable strategy for transitioning into sustainable farming. Particularly, legumes are considered important for their multiple benefits in the farming system such as sustaining soil nitrogen, reducing soil erosion, and increasing crop yield. Cover legumes have been recommended as a feasible strategy for smallholder coffee farmers. Similarly, it has been demonstrated that cover crop mixtures increase biomass inputs in the soil (Tibasiima et al, 2023).

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The decision to cultivate a cover crop is heavily influenced by the farmer's goals, such as preventing soil erosion, providing fertility, insect control, yield enhancement, and suppressing root-nematodes in cropping systems (Barry, 2020). Intercropping is being promoted as one of the integrated soil fertility management strategies that involves growing two or more crops in the same area at the same time. It has been used for decades and has helped agriculture achieve its goals. In addition, intercropping systems are beneficial to smallholder farmers in the low-input and/or high-risk environment, where intercropping of coffee and legumes is widespread among smallholder farmers due to the ability of the legume to contribute to addressing the problem of climate change and declining levels of soil fertility. The principal reasons for smallholder farmers to intercrop with legumes are flexibility, profit maximization, risk minimization, soil conservation and improvement of soil fertility, weed, pest, and disease control, and balanced nutrition (Lusembo et al., 2019).

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The objective of the study is to develop a micropropagation protocol on Robusta coffee through shoot tip cultures. Specifically, it aims to:

1. evaluate the nodulation, growth and yield responses of leguminous crops (soybean, peanut and mung bean) applied with plant growth regulators (PGRs) under coffee-based farming system;
2. determine which PGRs will give the better growth and yield on leguminous crops; and
3. evaluate which PGRs will give the better nodulation on leguminous crops.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Importance and Botanical Description

Coffee is one of the most important crops worldwide. Coffee species grow positively between 800 to 2000 m above sea level in equatorial and tropical climates (Jaramillo, 2005). Coffee is native to Africa, and belongs to the Rubiaceae family (Farah & Dos Santos, 2015). Global production of coffee achieved 10.5 million t, while in Mexico it was 240,000 t (. International Coffee Organization, 2023). The coffee species arabica (*Coffea arabica*) and robusta (*Coffea canephora* var. Robusta) account for 70% and 30% of global production, respectively (Davis & Mannii, 2011).

It is one of the most economic important crops for more than 50 countries in the world and it is highly valuable for beverage consumption. Out of more than 100 coffee species, only *Coffea arabica* L. (Arabica type coffee) and *C. canephora* Pierre (Robusta type coffee) are commercial species (Etienne 2006). Coffee is an important beverage crop owing to its economic benefits at individual and national levels (<https://www.ncausa.org/About-Coffee/What-is-Coffee>).

Coffea canephora is the commercially most important coffee variety. Out of it, Robusta Coffee is gained. The seeds of *Coffea canephora* are very rich in caffeine. They contain up to 4% of the stimulating substance. In comparison, Arabica Coffee only contains 1-2%. Robusta Coffee plants are - like the name already tells - extraordinarily resistant and are often cultivated in West Africa and East Asia. In the Mediterranean countries, coffee mixtures with an amount of Robusta are very popular. Many Espresso lovers think that the caffeine-rich Robusta bean gives the coffee the right spice ([https://www.magicgardenseeds.com/The-Unique/Robusta-Coffee-\(Coffea-canephora\)-A.COF03-](https://www.magicgardenseeds.com/The-Unique/Robusta-Coffee-(Coffea-canephora)-A.COF03-)).

Intercropping of Leguminous Crops to Coffee

Most researchers believe that the coffee legume intercrop system is beneficial to smallholder farmers in the low-input/high-risk environment of the tropics (Negi et al.2011). Intercropping of coffee and legumes is widespread among smallholder farmers due to the ability of the legume to cope with high temperatures and climate change, soil erosion control with declining levels of soil fertility. The principal reasons for smallholder farmers to intercrop coffee with legumes are flexibility, profit maximization, risk minimization against total crop failure, soil conservation and improvement of soil fertility, weed control, and balanced nutrition (Okoth and Siameto, 2011).

Furthermore, intercropping systems can give higher coffee yields than sole crop yields; greater yield stability; more efficient use of nutrients; better weed control; provision of insurance against total crop

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failure; and improved coffee quality. Also, crop legume as a sole crop requires a larger area to produce the same yield as a legume in an intercropping system (Obadoni et al., 2010).

Dahmardeh et al. (2010) reported that coffee Indigofera intercropping increases the amount of nitrogen, phosphorus, and potassium content compared to monocrops of coffee. Despite the beneficial effects of intercropping on cereal crops, it may also accelerate soil nutrient depletion, particularly for phosphorous, due to more efficient use of soil nutrients and higher removal through the harvested crops (Mucheru-Muna, et al., 2010).

Dahmardeh et al. (2010) reported that coffee Indigofera intercropping increases the amount of nitrogen, phosphorus, and potassium content compared to monocrops of coffee. Despite the beneficial effects of intercropping on cereal crops, it may also accelerate soil nutrient depletion, particularly for phosphorous, due to more efficient use of soil nutrients and higher removal through the harvested crops (Mucheru-Muna, et al., 2010).

Coffee-legume Intercrop systems control soil erosion by preventing rain drops from hitting the bare soil where they tend to seal surface pores, prevent water from entering the soil, and increase surface runoff (Seran and Brintha, 2010). In the coffee-Indigofera spicata intercropping system, spicata acted as the best cover crop and reduced soil erosion more than the coffee-bean system (Biasutti et al., 2009). In intercrops of taller coffee with short legume crops (Reddy and Reddi, 2007), taller crops act as wind barriers for shorter crops. Similarly, sorghum-cowpea intercropping reduced runoff by 20-30% compared to sorghum as a single crop and by 45-55 % compared to cowpea monoculture. Furthermore, intercropping reduced soil loss by more than 50% when compared to sorghum and cowpea monocultures (Zougmore et al., 2000).

Effect of Plant Growth Regulators to Soybean (*Glycine max* L.) Crops

Plant growth regulators when applied in very small quantity influence the plant growth. Several reports indicated that application of growth regulator improved the plant growth and yield. GA₃ enlarged length of stem and flower number plant-1 (Lee et al., 1999). GA₃ accelerated stem elongation and bud development (Kabar, 1990). Kinetin increased the fresh weight by increasing stem diameter in morning glory but reduced shoot length (Chaudhry and Khan, 2000). Salicylic acid is an endogenous signaling molecule and it has several functions, particularly the suppression of germination and growth, intervention with root absorption, reduced leaf abscission and transpiration (Ashraf et al., 2010; Hayat et al., 2010). Application of salicylic acid also significantly increased root dry weight. Salicylic acid application to soybean and corn promoted dry weight and leaf area of plants (Khan et al., 2003). The use of growth regulators is becoming popular to increase crop yield and varieties of such substances are available in the market which is being utilized for crop production.

The potential yield of the soybean crop is mainly influenced by the production, fixing and transformation of the reproductive structures into pods with the seeds, thus the spatial dynamics of these structures need to be studied and correlated with yield (Pires et al., 2000). Reports so far been made to indicate a promising result on yield of soybean and other oil crops due to the use of bio-chemical substances or plant growth regulator such as salicylic acid, gibberellic acid (GA₃), kinetin etc.

Effect of Plant Growth Regulators to Peanut (*Arachis hypogaea* L.) Crops

Growth hormones can contribute to the improved growth and development of field crops with higher yields, e.g. groundnut (Khan et al. 2011). In addition, auxins and GA₃ are important growth hormones that can affect the development of plant morphological and yield contributing attributes. Besides their primary role in growth regulation, growth hormones also affect timing of flowering, the sex of the flowers, senescence of leaves and fruits, fruit development, etc. Most of the workers have studied the effect of growth regulators on seed germination.

Plant growth regulators are organic compounds which in small amount modify a given physiological process. A number of studies have demonstrated the effect of plant growth regulators in altering several physiological parameters and hence the yield of crops. Menon and Srivastava (1984) emphasized the importance of plant growth regulators in source and sink relationships leading to enhanced translocation

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of photosynthates. Plant growth regulators such as gibberellic acid control some processes in plant growth and development such as seed germination in soybean, common bean, faba bean and cotton plants (Maske et al., 1997). It also controls root development like in the prevention or promotion of root elongation, and flowering and fruit development (Yamaguchi and Kamiya, 2000).

Effect of Plant Growth Regulators to Mungbean (*Vigna radiata* L.) Crops

Gibberellic acid (GA) has been used to increase the length or height of mung bean plants, increase the number of flowers and induce early flowering. Gibberellins play a major role in root growth, induce mitosis in the leaves, increase seed germination rate, the control of flowering time even organ elongation (Giri et al. 2018) the use of GA₃ greatly enhances the growth of plants and total area of leaf surfaces and also prevents premature falling of fruits. The application of Naphthalene acetic acid (NAA) can increase fruit setting ratio, prevent fruit dropping, promote flower sex ratio. Foliar application of NAA has also found to increase plant height, number of leaves per plant, fruit size with consequent enhancement in seed yield in different crops (Parmar et al. 2012).

Salicylic acid is a growth regulator of phenolic nature, which participates in the regulation of physiological processes in plant, such as stomata! closure, ion uptake, inhibition of ethylene biosynthesis, transpiration and stress tolerance. Foliar application of salicylic acid exerted a significant effect on plant growth metabolism when applied at physiological concentration and thus acted as one of the plant growth regulating substances (Rowndel et al. 2019).

TIBA (2, 3, 5-triiodo benzoic acid), a synthetic growth regulator is used for checking the excessive vegetative growth and lodging tendency, reducing the abscission of flowers and immature pods and for modification of crop canopy to improve the productivity of crops (Adam and Jahan 2014). Cycocel is known at sufficient concentration to counteract the effect of indigenous ABA and reduce the percentage of shedding (Singh et al., 2017). Other bio-regulators as TRIA has been reported to have profound effect on increasing the seed weight.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

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(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Study I. Nodulation, Growth, and Yield Responses of Soybean Applied with Plant Growth Regulators (PGRs) Under Coffee- Based Farming System

Experimental Design and Treatments

The study will be arranged in Randomized Complete Block Design (RCBD) with three replications. There will be 10 sample explants per replication. The experiment will be composed of four plant growth regulators namely: T₁ = control (water), T₂ = Gibberellic acid (50 ppm), T₃ = Gibberellic acid (100 ppm), T₄ = Gibberellic acid (150 ppm), T₅ = Salicylic acid (25 ppm), T₆ = Salicylic acid (50 ppm), and T₇ = Salicylic acid (50 ppm).

Collection of soil sample for analysis. Prior to the conduct of the study, soil samples will be collected from the coffee demonstration farm area. Composite soil sample of one kilogram will be placed in a properly labelled plastic bag and brought to the CAF Laboratory Building for air drying. Once air dried, the soil sample will be pulverized and sieve using 2-mm mesh and will be brought to any available soil's laboratory for analysis. This will be served as initial analysis for the study.

Land preparation. The experimental area of 18 m x 8 meters (144 m²) will be cleared using bolo. Unnecessary debris will be removed from the experimental area to facilitate easy cultivation. After the final harrowing, construction of drainage canals will be made.

Planting material. Soybean seeds (NSIC Sy 11 variety) will be procured from Bureau of Plant Industry in La Granja Experiment Station, La Carlota City, Negros Occidental.

Planting. This will be done as soon as the furrows are ready. The seeds will be sown directly into the plots at two seeds per hill following the recommended planting distance of 50 cm between rows and 10 cm between hill. The seeds will be covered with fine soil at approximately 2 to 3 cm deep to maintain uniform growth of seedlings.

Thinning and replanting. This will be done seven days after planting (DAP) by uprooting the unhealthy and less vigorous plant and leaving only one plant per hill. Replanting will be done using the healthy border plants.

Preparation of plant growth regulators (PGRs). For treatment 2, 50 ppm stock solution will be prepared by weighing 50 mg (0.05 g) of GA3 and will be diluted in 1 liter of distilled water. For treatment 3 and 4, 100 ppm and 150 ppm stock solution will be prepared by weighing 100 mg (0.1 g) and 150 mg (1.05 g) and will be diluted in 1 liter of water, respectively. For treatment 5, 25 ppm stock solution will be prepared by weighing 25 mg (0.025 g) and will be diluted in 1 liter of distilled water. For treatment 5 and 6, 50 ppm and 75 ppm stock solution will be prepared by weighing 50 mg (0.05 g) and 75 mg (0.075 g) and will be diluted in 1 liter of water, respectively.

Application of plant growth regulators (PGRs). Spraying of PGRs will be done at vegetative stage at 25 Days after sowing (DAS), flower initiation stage at 40 DAS and pod initiation stage at 50 DAS.

Application of synthetic fertilizer. Application of synthetic fertilizer will be done following the recommended rate after the result of the soil analysis.

Weed management. Weed control will be done hand weeding method. This will be done when weeds will be observed in the experimental area to avoid competition with nutrients, sunlight and space.

Water management. Watering of plants will be done very other day or when the need arises.

Prevention and control of pest and diseases. Maintaining cleanliness and sanitation in the experimental

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area will be done. Oriental Herbal Nutrient (OHN) will be applied to control insect pests and diseases that will be observed in the experimental area.

Harvesting. The soybean pods will be harvested between 60-90 DAP. This will be done by carefully picking the pods from the plants. Harvested pods will be partially dried to facilitate ease during threshing operation. This will be done by separating the pods from grains using feet trampling. The mature pods will be separated from immature pods.

Nodules

Number of nodules. This will be determined by carefully uprooting the roots and nodules will be collected and counted.

Number of active and inactive nodules. This will be done by cutting the nodules into half. Active nodules will be determined by its reddish to pinkish color, while the inactive will be determined by its greenish to black color.

Growth Parameters

Plant Height (cm). This will be taken from the base of the sample plants to the tip of the tallest leaf using tape measure.

Biomass (g). This will be done by weighing the whole sample plants after removing the soil adhering to the roots.

Root weight (g). This will be determined by weighing the roots of the sample plants.

Root length (cm). This will be done by measuring the length of the roots using tape measure.

Yield Parameters

Number of pods. This will be determined by counting the number of pods of the sample plants.

Number of seeds per pod. From the 10 sample plants in each plot, 10 pods will be selected and the number of seeds each pod will be counted.

Pod length (cm). This will be done by measuring the pod length using a tape measure.

Seed weight (kg/ha). The seeds collected from 4 (2 m × 2 m) square meter of each plot will be cleaned. The weight of seeds was taken and converted the yield in kg ha⁻¹.

Stover yield (kg/ha). Stover obtained from each unit plot will be sun-dried and weighed carefully. The dry weight of stover of soybean 1 m² area and five sample plants will be added to the respective stover yield m² and finally converted to kg ha⁻¹.

Biological yield (kg/ha). Seed yield and stover yield together will be regarded as biological yield. The biological yield will be calculated with the following formula:

$$\text{Biological yield} = \text{Seed yield} + \text{Stover yield}$$

Harvest index (%). Harvest index will be calculated from the seed and stover yield of soybean for each plot and expressed in percentage.

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Economic yield (seed weight)

$$HI = \frac{\text{Biological yield}}{\text{Biological yield}} \times 100$$

Moisture and Protein Percentage. After collecting the seed, the seed samples will be sun dried and then packed in polythene bag by proper labeling. These labeled packed samples will be immediately sent to any available testing center for determination of moisture percentage and crude protein percentage.

Study II. Nodulation, Growth, and Yield Responses of Peanut Applied with Plant Growth Regulators (PGRs) Under Coffee- Based Farming System

Experimental Design and Treatments

The study will be arranged in Randomized Complete Block Design (RCBD) with three replications. There will be 10 sample explants per replication. The experiment will be composed of four plant growth regulators namely: T₁ = control (water), T₂ = Gibberellic acid (50 ppm), T₃ = Gibberellic acid (100 ppm), T₄ = Gibberellic acid (150 ppm), T₅ = Salicylic acid (25 ppm), T₆ = Salicylic acid (50 ppm), and T₇ = Salicylic acid (50 ppm).

Collection of soil sample for analysis. Prior to the conduct of the study, soil samples will be collected from the coffee demonstration farm area. Composite soil sample of one kilogram will be placed in a properly labelled plastic bag and brought to the CAF Laboratory Building for air drying. Once air dried, the soil sample will be pulverized and sieve using 2-mm mesh and will be brought to any available soil's laboratory for analysis. This will be served as initial analysis for the study.

Land preparation. The experimental area of 18 m x 8 meters (144 m²) will be cleared using bolo. Unnecessary debris will be removed from the experimental area to facilitate easy cultivation. After the final harrowing, construction of drainage canals will be made.

Planting material. Peanut seeds (NSIC Pn 18 variety) will be procured from Bureau of Plant Industry in La Granja Experiment Station, La Carlota City, Negros Occidental.

Planting. This will be done as soon as the furrows are ready. The seeds will be sown directly into the plots at two seeds per hill following the recommended planting distance of 50 cm between rows and 10 cm between hill. The seeds will be covered with fine soil at approximately 2 to 3 cm deep to maintain uniform growth of seedlings.

Thinning and replanting. This will be done seven days after planting (DAP) by uprooting the unhealthy and less vigorous plant and leaving only one plant per hill. Replanting will be done using the healthy border plants.

Preparation of plant growth regulators (PGRs). For treatment 2, 50 ppm stock solution will be prepared by weighing 50 mg (0.05 g) of GA₃ and will be diluted in 1 liter of distilled water. For treatment 3 and 4, 100 ppm and 150 ppm stock solution will be prepared by weighing 100 mg (0.1 g) and 150 mg (1.05 g) and will be diluted in 1 liter of water, respectively. For treatment 5, 25 ppm stock solution will be prepared by weighing 25 mg (0.025 g) and will be diluted in 1 liter of distilled water. For treatment 5 and 6, 50 ppm and 75 ppm stock solution will be prepared by weighing 50 mg (0.05 g) and 75 mg (0.075 g) and will be diluted in 1 liter of water, respectively.

Application of plant growth regulators (PGRs). Spraying of PGRs will be done at vegetative stage at 25 Days after sowing (DAS), flower initiation stage at 40 DAS and pod initiation stage at 50 DAS.

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Application of synthetic fertilizer. Application of synthetic fertilizer will be done following the recommended rate after the result of the soil analysis.

Weed management. Weed control will be done hand weeding method. This will be done when weeds will be observed in the experimental area to avoid competition with nutrients, sunlight and space.

Water management. Watering of plants will be done very other day or when the need arises.

Prevention and control of pest and diseases. Maintaining cleanliness and sanitation in the experimental area will be done. Oriental Herbal Nutrient (OHN) will be applied to control insect pests and diseases that will be observed in the experimental area.

Harvesting. The soybean pods will be harvested between 60-90 DAP. This will be done by carefully picking the pods from the plants. Harvested pods will be partially dried to facilitate ease during threshing operation. This will be done by separating the pods from grains using feet trampling. The mature pods will be separated from immature pods.

Nodules

Number of nodules. This will be determined by carefully uprooting the roots and nodules will be collected and counted.

Number of active and inactive nodules. This will be done by cutting the nodules into half. Active nodules will be determined by its reddish to pinkish color, while the inactive will be determined by its greenish to black color.

Growth Parameters

Plant Height (cm). This will be taken from the base of the sample plants to the tip of the tallest leaf using tape measure.

Biomass (g). This will be done by weighing the whole sample plants after removing the soil adhering to the roots.

Root weight (g). This will be determined by weighing the roots of the sample plants.

Root length (cm). This will be done by measuring the length of the roots using tape measure.

Yield Parameters

Number of pods. This will be determined by counting the number of pods of the sample plants.

Number of seeds per pod. From the 10 sample plants in each plot, 10 pods will be selected and the number of seeds each pod will be counted.

Pod length (cm). This will be done by measuring the pod length using a tape measure.

Seed weight (kg/ha). The seeds collected from 4 (2 m × 2 m) square meter of each plot will be cleaned. The weight of seeds was taken and converted the yield in kg ha⁻¹.

Stover yield (kg/ha). Stover obtained from each unit plot will be sun-dried and weighed carefully. The dry weight of stover of soybean 1 m² area and five sample plants will be added to the respective stover yield m² and finally converted to kg ha⁻¹.

Biological yield (kg/ha). Seed yield and stover yield together will be regarded as biological yield. The

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biological yield will be calculated with the following formula:

$$\text{Biological yield} = \text{Seed yield} + \text{Stover yield}$$

Harvest index (%). Harvest index will be calculated from the seed and stover yield of soybean for each plot and expressed in percentage.

$$\text{HI} = \frac{\text{Economic yield (seed weight)}}{\text{Biological yield}} \times 100$$

Moisture and Protein Percentage. After collecting the seed, the seed samples will be sun dried and then packed in polythene bag by proper labeling. These labeled packed samples will be immediately sent to any available testing center for determination of moisture percentage and crude protein percentage.

Study III. Nodulation, Growth, and Yield Responses of Mung bean Applied with Plant Growth Regulators (PGRs) Under Coffee- Based Farming System

Experimental Design and Treatments

The study will be arranged in Randomized Complete Block Design (RCBD) with three replications. There will be 10 sample explants per replication. The experiment will be composed of four plant growth regulators namely: T₁ = control (water), T₂ = Gibberellic acid (50 ppm), T₃ = Gibberellic acid (100 ppm), T₄ = Gibberellic acid (150 ppm), T₅ = Salicylic acid (25 ppm), T₆ = Salicylic acid (50 ppm), and T₇ = Salicylic acid (50 ppm).

Collection of soil sample for analysis. Prior to the conduct of the study, soil samples will be collected from the coffee demonstration farm area. Composite soil sample of one kilogram will be placed in a properly labelled plastic bag and brought to the CAF Laboratory Building for air drying. Once air dried, the soil sample will be pulverized and sieve using 2-mm mesh and will be brought to any available soil's laboratory for analysis. This will be served as initial analysis for the study.

Land preparation. The experimental area of 18 m x 8 meters (144 m²) will be cleared using bolo. Unnecessary debris will be removed from the experimental area to facilitate easy cultivation. After the final harrowing, construction of drainage canals will be made.

Planting material. Mung bean seeds (NSIC Mg 18 variety) will be procured from Bureau of Plant Industry in La Granja Experiment Station, La Carlota City, Negros Occidental.

Planting. This will be done as soon as the furrows are ready. The seeds will be sown directly into the plots at two seeds per hill following the recommended planting distance of 50 cm between rows and 10 cm between hill. The seeds will be covered with fine soil at approximately 2 to 3 cm deep to maintain uniform growth of seedlings.

Thinning and replanting. This will be done seven days after planting (DAP) by uprooting the unhealthy and less vigorous plant and leaving only one plant per hill. Replanting will be done using the healthy border plants.

Preparation of plant growth regulators (PGRs). For treatment 2, 50 ppm stock solution will be prepared by weighing 50 mg (0.05 g) of GA3 and will be diluted in 1 liter of distilled water. For treatment 3 and 4, 100 ppm and 150 ppm stock solution will be prepared by weighing 100 mg (0.1 g) and 150 mg (1.05 g) and will be diluted in 1 liter of water, respectively. For treatment 5, 25 ppm stock solution will be prepared

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by weighing 25 mg (0.025 g) and will be diluted in 1 liter of distilled water. For treatment 5 and 6, 50 ppm and 75 ppm stock solution will be prepared by weighing 50 mg (0.05 g) and 75 mg (0.075 g) and will be diluted in 1 liter of water, respectively.

Application of plant growth regulators (PGRs). Spraying of PGRs will be done at vegetative stage at 25 Days after sowing (DAS), flower initiation stage at 40 DAS and pod initiation stage at 50 DAS.

Application of synthetic fertilizer. Application of synthetic fertilizer will be done following the recommended rate after the result of the soil analysis.

Weed management. Weed control will be done hand weeding method. This will be done when weeds will be observed in the experimental area to avoid competition with nutrients, sunlight and space.

Water management. Watering of plants will be done very other day or when the need arises.

Prevention and control of pest and diseases. Maintaining cleanliness and sanitation in the experimental area will be done. Oriental Herbal Nutrient (OHN) will be applied to control insect pests and diseases that will be observed in the experimental area.

Harvesting. The soybean pods will be harvested between 60-90 DAP. This will be done by carefully picking the pods from the plants. Harvested pods will be partially dried to facilitate ease during threshing operation. This will be done by separating the pods from grains using feet trampling. The mature pods will be separated from immature pods.

Nodules

Number of nodules. This will be determined by carefully uprooting the roots and nodules will be collected and counted.

Number of active and inactive nodules. This will be done by cutting the nodules into half. Active nodules will be determined by its reddish to pinkish color, while the inactive will be determined by its greenish to black color.

Growth Parameters

Plant Height (cm). This will be taken from the base of the sample plants to the tip of the tallest leaf using tape measure.

Biomass (g). This will be done by weighing the whole sample plants after removing the soil adhering to the roots.

Root weight (g). This will be determined by weighing the roots of the sample plants.

Root length (cm). This will be done by measuring the length of the roots using tape measure.

Yield Parameters

Number of pods. This will be determined by counting the number of pods of the sample plants.

Number of seeds per pod. From the 10 sample plants in each plot, 10 pods will be selected and the number of seeds each pod will be counted.

Pod length (cm). This will be done by measuring the pod length using a tape measure.

Seed weight (kg/ha). The seeds collected from 4 (2 m × 2 m) square meter of each plot will be cleaned. The weight of seeds was taken and converted the yield in kg ha⁻¹.

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Stover yield (kg/ha). Stover obtained from each unit plot will be sun-dried and weighed carefully. The dry weight of stover of soybean 1 m² area and five sample plants will be added to the respective stover yield m² and finally converted to kg ha-1.

Biological yield (kg/ha). Seed yield and stover yield together will be regarded as biological yield. The biological yield will be calculated with the following formula:

$$\text{Biological yield} = \text{Seed yield} + \text{Stover yield}$$

Harvest index (%). Harvest index will be calculated from the seed and stover yield of soybean for each plot and expressed in percentage.

$$\text{HI} = \frac{\text{Economic yield (seed weight)}}{\text{Biological yield}} \times 100$$

Moisture and Protein Percentage. After collecting the seed, the seed samples will be sun dried and then packed in polythene bag by proper labeling. These labeled packed samples will be immediately sent to any available testing center for determination of moisture percentage and crude protein percentage.

Data Analysis Procedures

All sets of data that will be gathered from the study will be analyzed using the Analysis of Variance (ANOVA). The significant differences among treatment means will be determined using the Duncan Multiple Range Test (DMRT). STAR software will be used in analyzing the data gathered in the study.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1			Year 2	
	Q2	Q3	Q4	Q1	Q2
1. Collection and Submission of Soil Sample					
2. Land Preparation					
3. Planting					
4. Data Collection					
5. Data Analysis and Interpretation					
6. Analysis for Moisture and Protein Percentage					

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

- Patent/Intellectual Property-** application of PGRs, 3 IECs
- People Services-** trainings/demo with potential farmers
- Partnership-** Funding Agencies/ Farmer's Association

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(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities) 1. application of PGRs to leguminous crops intercropped with coffee			
(17) Target Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output) 1. Potential agronomic crop farmers/growers 2. Farmer's Association			

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(18) Personnel Requirement		
Name	Role (Program, Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement: 50, 000.00

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		
- Meal allowance during data gathering		
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		
- Printing		
C. CAPITAL OUTLAY		
Total		

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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(21) Capsule Curriculum Vitae (One-page brief C.V. only)					